

KaarigarAI — 9-Day Build Roadmap

The AI Virtual Business Manager for Marginalized Artisans · SIH26090 · Team Algorhythm. Nine days to a working prototype and a rehearsed demo. Docs (proposal, deck, architecture, wireframes) are already revised — this covers the build.

The premise, stated honestly

The three mandated features — image enhancer, multilingual auto-cataloger, dynamic pricing — are what get scored. Everything after them is the market-linkage half the problem statement asks for and most submissions forget. Nine days is enough for a **near-full-time sprint**, not a part-time one. Two ordering rules decide the whole plan:

- **The shared text-to-speech voice is built on Day 1**, because read-back confirmation, onboarding, spoken alerts and quality nudges all depend on it.
- **The public listing page is built before the offer inbox, and the storefront after it** — a buyer needs somewhere to send an offer from, and the storefront is aggregation over pages the listing work already produces.

Scope is locked by five decisions already made: **no buyer marketplace** (permanent listing page + storefront + direct offer instead), **no material classifier** (material comes from her voice, the photo only suggests category/size/finish), **dated + sourced material rates on screen**, **listings carry stock** (unique or batch), and a **per-artisan storefront** as the year-round channel.

Non-negotiable gates (do not skip under time pressure)

Gate	Day	If it fails
Photo + voice note → bilingual listing, read back aloud , on a real device	3	The low-literacy claim is the whole product. Fix the pipeline before pricing.
The floor visibly refuses an underpriced suggestion; material is never inferred from the photo	4	Pricing logic is wrong — stop and fix before building on it.
A batch listing sells down with no double-accept ; artisan accepts by voice	5	The stock/offer transaction is wrong. Don't demo it.
Full seven-beat demo runs twice with no internet	7	Record the video now and stop polishing the live path.

The long pole: cloud steps and free-tier caps

Transcription, translation, description and pricing publish are network steps, and the free tiers (Gemini especially) carry daily caps. **Pre-cache every response for the demo items in advance**, and keep the on-device speech model as the offline fallback. Never call any API inside a training or batch loop.

Emphasis legend: ▲ what gets scored · ★ differentiator (protect the hours) · △ highest risk · ○ supporting

Day by day

Day 1 — Image pipeline + the shared voice ▲ △(dependency)

- **AI Image Enhancer** (mandated 1): rembg (U2-Net) background removal, OpenCV white-balance + histogram + saliency crop, **failure-aware retake** (keep original on a damaged cut-out), texture close-up for textiles. Add blur/shake detection at capture if time.
- **Shared TTS voice** — built here because everything downstream speaks. △ If this slips, read-back, onboarding and nudges all slip with it.
- **Exit:** a raw phone photo → clean e-commerce image on a real device; TTS speaks a string aloud.

Day 2 — Voice → listing ▲

- **Multilingual Auto-Cataloger** (mandated 2): whisper.cpp transcription (server default, on-device fallback), IndicTrans2 translation, Gemini bilingual description.
- Craft-glossary fuzzy-correct · **PI-strip** (identifier-like digit runs) · low-confidence re-record · **spoken read-back confirmation** — nothing published without it.
- **Exit:** a regional-language voice note → bilingual listing, read back aloud, confirmed by voice.

Day 3 — Flutter shell + offline queue · GATE

- Camera capture, voice record, local SQLite, **outbound queue** (client-generated ids, image downscale, oldest-first sync, size cap), draft autosave, queued-state UI.
- Server-side dedupe so a repeated push can't duplicate a listing.
- **Exit gate:** capture + record + queue works **offline** on device and syncs with no duplicate; and the Day-2 pipeline runs end to end on a real device.

Day 4 — Dynamic pricing ▲ · GATE

- **Attributes:** vision proposes category / size class / finish as a *suggestion*; **material comes from the voice note**; 0–2 spoken confirmations. **No material classifier is trained.**
- Material cost asked **once per craft type** (stored against her); **dated, sourced material-rate table** as fallback, printed on screen.
- XGBoost price **band** · SHAP three bars · **comparables evidence strip** (3 similar listings + her ~cost) shown *before* the model number · **fair-price floor** (warn, never block) · bounded seasonal multiplier · out-of-range honesty (wider band, stated low confidence).
- **Exit gate:** photo + voice → price band with a plain explanation, the floor **visibly refusing** an underpriced suggestion, and the rate's **date + source on screen.**

Day 5 — Onboarding + listing page + offer/stock ▲ · GATE

- **Voice-first onboarding:** language tiles that speak their own name, numeric keypad, OTP read aloud, **deferred verification** (unverified capture allowed, publish blocked).
- **Permanent public listing page. Offer inbox:** structured offer (price / quantity / date / contact), asking-price **auto-decline**, quantity validated against remaining count, **unique-vs-batch stock**, decrement under a row lock inside the acceptance transaction, sold-out state + **spoken restock**, expiry. Accept/decline **by voice**. Report-an-issue record.
- **Exit gate:** a buyer opens the listing page, offers for N of a batch, artisan accepts by voice, remaining count drops — no double-accept.

Day 6 — Storefront + market linkage ★ — the half most forget

- **Artisan storefront** (permanent per-artisan URL): maker story on top, every live listing below, sold-out marked. **Follow** (email only, weekly batched digest, one-click unsubscribe), **returning-buyer flag** spoken with the offer, follower count on the dashboard.
- **Stall QR → storefront. Share card** (per-listing QR + storefront address). **Export bundle** (GeM / ONDC / Amazon Karigar / ODOP). **Maker story** pipeline (reuses Day-2 voice pipeline).
- ◦ Spoken **scheme / mela / seasonal alerts · quality meter** + icon-driven **dashboard** (listings, offers, monthly earning, remaining stock, followers) · **voice order & status** (intent matcher over a few fixed questions).
- **Exit:** stall QR opens the storefront; one tap yields a valid export bundle; dashboard shows real aggregates.

Day 7 — Edge-case hardening + offline rehearsal + pre-cache △ · GATE

- Harden the register that causes visible failure: noisy audio → re-record · mis-heard craft word → glossary correct · PI digits stripped · pale-on-pale → retake · unseen category → wider band · per-piece vs per-set → ask · two buyers / one batch → locked decrement · no offers for a week → reasoned nudge · growing offline queue → cap + downscale + oldest-first · middleman → account & QR bound to her own number, earnings only in her view.
- **Offline rehearsal** with the network disabled; **pre-cache** Gemini + TTS responses for every demo item.
- **Exit gate:** the full seven-beat demo runs **twice with no internet**.

Day 8 — Freeze + seven-beat rehearsal + video

- **Freeze the build. No functional change after this point.** Tag it.
- Rehearse the seven beats: first run → photograph → voice note → price → offline → **the buyer** (scan stall QR → storefront → maker story → offer → accept by voice) → **linkage** (export bundle + follow; close on a real scheme).
- **Live linkage API for the demo only**, with the **cached fallback tested now, not on the day**.
- Record a **90-second backup video** of the demo working.
- **Exit gate:** seven-beat demo runs end to end, twice, offline; video recorded.

Day 9 — Final docs pass + submission + buffer

- Proposal, deck, architecture, wireframes are already revised — **final pass only:** README so the app runs from a clean clone, the demo script, and every section mapped to a problem-statement impact goal.
- **Lead with the measured story:** one photo + one voice note, in her own language, to a published listing and a buyer — the professional photographer, translator, copywriter and pricing analyst she could never afford to hire.
- Submit. Hold ~2 hrs as slack.

Fallback ladder (spend these in order if a day slips)

1. **Image extras first** — drop synthetic shadow / perspective de-skew / angle coach; keep background removal + correction + crop + failure-aware retake. The core is mandated; the extras

are not.

2. **Storefront before the follow** — ship the storefront (aggregation, cheap) and defer the email digest. The QR-opens-her-catalogue story survives without the digest.
3. **Demo on cached responses only**, skip the live linkage API — still shows the one decision.
4. **One tested language end to end** (Hindi) for the demo; describe the other three as supported-but-unverified rather than faking them live.

Never cut: the three mandated features working on a real device (Days 1–4), the offline capture + read-back (the low-literacy claim), and the storefront + offer (the market-linkage half).

Watch items

- **Free-tier caps + demo-day internet** — pre-cached responses and the on-device speech model are the mitigation. Decide and test the API fallback on Day 7, not on the day.
- **Permanent URL scheme is irreversible** once a stall QR is printed — freeze the artisan/listing id format deliberately (Day 5), not by accident later.
- **Pricing quality now rests on one spoken answer** (her material cost) — if it's misheard, the floor is wrong. Confirm it by voice.
- **Middleman capture** — bind account + stall QR to her own number, earnings visible only in her view.

Cut list — still out (from the proposal)

payment · shipping · **dispute resolution** (the report-an-issue record only logs) · full live auctions · WhatsApp inbound listing · cooperative / self-help-group storefront · peer benchmarking · multi-angle spin view · buyer accounts & order history · automated material-rate refresh · AI upscaling · the ~16 further languages the models support but that aren't tested.